

In-Memory Cache (LRU / LFU / FIFO)

Patterns: Strategy Pattern (Eviction), Decorator Pattern (Stats/TTL), Template Method | Domain: Infrastructure / High-Performance KV

Core Requirements & Features	Core Domain Classes & Interfaces
• Key-Value Store: Fast O(1) get(key) and put(key, value) operations. • Pluggable Eviction: Support LRU (Least Recently Used), LFU (Least Frequently Used), FIFO policies. • TTL & High Concurrency: Support Key Expiration and thread-safe operations.	• InMemoryCache: Generic KV cache manager delegating eviction to strategy. • EvictionPolicy: Interface implemented by LRUEvictionPolicy, LFUEvictionPolicy, etc. • Node: Doubly-linked list node holding key, value, frequency count. • ConcurrentHashMap: Stores key -> Node pointers for O(1) direct memory access.

Critical Execution Workflow & Method Trace

1. Client calls cache.get(key).
2. InMemoryCache looks up key in ConcurrentHashMap.
3. If key found & not expired: EvictionPolicy.keyAccessed(key) is invoked.
4. LRU Eviction: Moves corresponding Node to HEAD of Doubly-LinkedList.
5. LFU Eviction: Increments Node.frequency, moves Node from freq_K bucket list to freq_K+1 bucket list.
6. Client calls cache.put(key, val) when capacity is full:
7. EvictionPolicy.evictKey() identifies target key: LRU removes TAIL node; LFU removes node from lowest frequency bucket.
8. Target key removed from ConcurrentHashMap & Doubly-LinkedList; new key inserted at HEAD.

Concurrency & Thread Safety Mechanics	Design Trade-offs & Interview FAQs
• ReadWriteLock: ReentrantReadWriteLock allowing concurrent readers and exclusive writer during eviction. • Fine-Grained Locks: Segmented locking or Striped locks over cache buckets. • Atomic Eviction: Synchronized eviction block ensures Map and LinkedList state remain 100% consistent.	• LRU vs LFU: LRU is susceptible to cache scan pollution; LFU handles frequency better but requires memory for freq buckets. • TTL Expiration: Passive expiration (on get) vs Active background cleanup thread (sweep every N sec). • Generics: Type-safe template support.